

Q1) Solution
Here,

$$\text{Pass Band ripple } (\delta_p) = 0.005$$

$$\text{Stop band ripple } (\delta_s) = 0.001$$

we have,

$$\begin{aligned} \text{i) } A_s &= -20 \log \delta_s \text{ dB} \\ &= -20 \log (0.001) \text{ dB} \\ &= \underline{\underline{60 \text{ dB}}} \end{aligned}$$

$$\begin{aligned} \text{ii) } A_p &= -20 \log (1 - \delta_p) \text{ dB} \\ &= -20 \log (1 - 0.005) \text{ dB} \\ &= \underline{\underline{0.044 \text{ dB}}} \end{aligned}$$

Q2) Solution
Here,

$$H_a(s) = \frac{s+2}{(s+1)(s+3)} \quad - (i)$$

Case: I $\Rightarrow$ Using IIM method:

$$H_a(s) = \frac{s+2}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

$$= \frac{1}{2} \cdot \frac{1}{s+1} + \frac{1}{2} \cdot \frac{1}{s+3}$$

$$\text{So, } H_a(s) = \frac{1}{2(s+1)} + \frac{1}{2(s+3)}$$

For IIM,

$$H_a(s) = \frac{1}{s - p_1} \rightarrow H(z) = \frac{1}{1 - e^{p_1 T} z^{-1}}$$

$$\begin{aligned}
 H(z) &= \frac{1}{2} \cdot \frac{1}{1 - e^{-2T} z^{-1}} + \frac{1}{2} \cdot \frac{1}{1 - e^{-3T} z^{-1}} \\
 &= \frac{1}{2} \left[\frac{1}{1 - e^{-2T} z^{-1}} + \frac{1}{1 - e^{-3T} z^{-1}} \right] \\
 &= \frac{1}{2} \left[\frac{1 - e^{-3T} z^{-1} + 1 - e^{-2T} z^{-1}}{(1 - e^{-2T} z^{-1})(1 - e^{-3T} z^{-1})} \right] \\
 &= \frac{1}{2} \left[\frac{2 - (e^{-2T} + e^{-3T}) z^{-1}}{(1 - e^{-2T} z^{-1})(1 - e^{-3T} z^{-1})} \right] \\
 &= \frac{1 - \frac{1}{2}(e^{-2T} + e^{-3T}) z^{-1}}{(1 - e^{-2T} z^{-1})(1 - e^{-3T} z^{-1})} \\
 &= \frac{z^{-1} [z - \frac{1}{2}(e^{-2T} + e^{-3T})]}{z^2 (z - e^{-2T})(z - e^{-3T})} \\
 &= \frac{z [z - \frac{1}{2}(e^{-2T} + e^{-3T})]}{(z - e^{-2T})(z - e^{-3T})}
 \end{aligned}$$

Hence, $\rightarrow$ Zeros are located at: $z=0$ & $z = \frac{1}{2}(e^{-2T} + e^{-3T})$
 $\rightarrow$ Poles are located at: $z = e^{-2T}$ & $z = e^{-3T}$

Case: II: Matched Z-transform:

$$H_a(s) = \frac{\prod_{k=1}^m (s - z_k)}{\prod_{k=1}^n (s - p_k)} = H(z) = \frac{\prod_{k=1}^m (1 - e^{z_k T} z^{-1})}{\prod_{k=1}^n (1 - e^{p_k T} z^{-1})}$$

So eqn (i) becomes,

$$\begin{aligned}
 H(z) &= \frac{(1 - e^{-2T} z^{-1})}{(1 - e^{-2T} z^{-1})(1 - e^{-3T} z^{-1})} \\
 &= \frac{z(z - e^{-2T})}{(z - e^{-2T})(z - e^{-3T})}
 \end{aligned}$$

Hence, Zeros are located at: $z=0$ & $z=e^{-2T}$
 Poles are located at: $z=e^{-T}$ & $z=e^{-3T}$

Q3) Solution
 Here,

$$H(s) = \frac{s+0.1}{(s+0.1)^2 + 9} \quad ; T = 0.1 \text{ sec}$$

Case: $f \Rightarrow$ Matched Z-Transform:

$$\begin{aligned} H(s) &= \frac{(s+0.1)}{(s+0.1)^2 - (j3)^2} \\ &= \frac{(s+0.1)}{(s+0.1+3j)(s+0.1-3j)} \end{aligned}$$

$$= \frac{s+0.1}{[s+(0.1+3j)][s+(0.1-3j)]}$$

now,

$$\begin{aligned} H(z) &= \frac{1 - e^{-0.2T} z^{-1}}{[1 - e^{-(0.1+3j)T} z^{-1}][1 - e^{-(0.1-3j)T} z^{-1}]} \\ &= \frac{z(z - e^{-0.2T})}{[z - e^{-(0.1+3j)T}][z - e^{-(0.1-3j)T}]} \end{aligned}$$

At $T = 0.1 \text{ sec}$,

$$\begin{aligned} H(z) &= \frac{z(z - 0.99)}{[z - 0.99e^{0.3j}][z - 0.99e^{-0.3j}]} \\ &= \frac{z(z - 0.99)}{z^2 - 0.99e^{0.3j}z - 0.99e^{-0.3j}z + (0.99)^2} \end{aligned}$$

$$\Rightarrow H(z) = \frac{z(z-0.99)}{z^2 - 0.99z(e^{0.3j} + e^{-0.3j}) + 0.98}$$

$$= \frac{z(z-0.99)}{z^2 - 0.99z(1.0803) + 0.98}$$

$$= \frac{z(z-0.99)}{z^2 - 1.07z + 0.98}$$

Hence,

Zeros/Poles are located at: $z=0$ & $z=0.99$

~~Zeros~~ Poles are located at: $p_1 = -0.985 + 0.09j$
 $p_2 = -0.985 - 0.09j$

Case II $\Rightarrow$ IIm method:

$$H(s) = \frac{s+0.1}{(s+0.1)^2 + 3^2}$$

$$\text{We have, } \frac{s+a}{(s+a)^2 + b^2} \Rightarrow H(z) = \frac{1 - e^{-aT} \cos bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}}$$

Here, $a = 0.1$; $b = 3$ and $T = 0.1 \text{ sec}$

So,

$$H(z) = \frac{1 - e^{-0.1 \times 0.1} \cos(3 \times 0.1) z^{-1}}{1 - 2e^{-0.1 \times 0.1} \cos(3 \times 0.1) z^{-1} + e^{-2 \times 0.1 \times 0.1} z^{-2}}$$

$$= \frac{1 - 0.985z^{-1}}{1 - 1.892z^{-1} + 0.982z^{-2}} = \frac{z(z-0.985)}{(z^2 - 1.892z + 0.982)}$$

Hence:

Zeros are at: $z=0$ & $z=0.985$

Poles located at: $p_1 = -0.945 - j0.295$

$p_2 = -0.945 + j0.295$

Ques: III $\Rightarrow$ RLT Method:

100 marks

$$H(z) = H_A(s) \Big|_{s=\frac{z-1}{z+1}} = \frac{20\left(\frac{z-1}{z+1}\right)}{s^2 + 0.1s + 0.1} = \frac{20\left(\frac{z-1}{z+1}\right)}{\left(\frac{z-1}{z+1}\right)^2 + 0.1\left(\frac{z-1}{z+1}\right) + 0.1}$$

$$\Rightarrow H(z) = \frac{20\left(\frac{z-1}{z+1}\right) + 0.1}{\left(\frac{z-1}{z+1}\right)^2 + 0.1\left(\frac{z-1}{z+1}\right) + 0.1}$$

$$= \frac{20\left[\frac{z-1}{z+1}\right] + 0.1\left[\frac{z+1}{z+1}\right]}{\left[\frac{z-1}{z+1}\right]^2 + 0.1\left[\frac{z-1}{z+1}\right] + 0.1\left[\frac{z+1}{z+1}\right]}$$

$$= \frac{20(z-1) + 0.1(z+1)}{(z+1)^2}$$

$$= \frac{(20z - 20 + 0.1z + 0.1)}{(z+1)^2} = \frac{(20.1z - 19.9)}{(z+1)^2}$$

$$= \frac{20.1z^2 - 800z + 400 + 40.2z - 4 + 0.02z + 0.01}{924 + 182z + 1}$$

$$\therefore H(z) = \frac{(z-1)(z+0.9)}{(z-(-0.95-j0.3))(z-(-0.95+j0.3))}$$

So, poles located at: $z = -1$ and $z = -0.95$
 poles located at: $p_1 = -0.95 - j0.3$
 $p_2 = -0.95 + j0.3$

(Q4) Calculation

$$H_c(s) = \frac{s+9}{(s+9)^2 + b^2}$$

$$\frac{s+9}{(s+9)^2 + b^2} = \frac{s+9}{(s+9+jb)(s+9-jb)} = \frac{A}{(s+9+jb)} + \frac{B}{(s+9-jb)}$$

$$\Rightarrow A = \frac{-a-jb+a}{-a-jb-jb} = \frac{-jb}{-2jb} = \frac{0.5}{2}$$

$$B = \frac{-a+jb+a}{-a+jb-jb} = \frac{jb}{-2jb} = \frac{0.5}{-2}$$

$$\Rightarrow H_c(s) = \frac{0.5}{(s+9+jb)} + \frac{0.5}{(s+9-jb)}$$

II. $h_c(t) = 0.5e^{-(a+jb)t} u(t) + 0.5e^{-(a-jb)t} u(t)$

$$= 0.5 \left[e^{-(a+jb)t} + e^{-(a-jb)t} \right] u(t)$$

Note $h_d(n) = h_c(nT) = 0.5 \left[e^{-(a+jb)nT} + e^{-(a-jb)nT} \right] u(n)$

$$H(z) = \sum_{n=0}^{\infty} h_d(n) z^{-n} = \sum_{n=0}^{\infty} 0.5 \left[e^{-(a+jb)nT} + e^{-(a-jb)nT} \right] z^{-n}$$

$$= \sum_{n=0}^{\infty} 0.5 e^{-(a+jb)nT} z^{-n} + \sum_{n=0}^{\infty} 0.5 e^{-(a-jb)nT} z^{-n}$$

$$= 0.5 \sum_{n=0}^{\infty} \left[e^{-(a+jb)nT} z^{-n} \right] + 0.5 \sum_{n=0}^{\infty} \left[e^{-(a-jb)nT} z^{-n} \right]$$

$$\begin{aligned}
 &= 0.5 \times \frac{1}{1 - e^{-(a+1b)} z^{-1}} + 0.5 \times \frac{1}{1 - e^{(a+1b)} z^{-1}} \\
 &= 0.5 \frac{1 - e^{-(a+1b)} z^{-1} + 1 - e^{(a+1b)} z^{-1}}{[1 - e^{-(a+1b)} z^{-1}][1 - e^{(a+1b)} z^{-1}]} \\
 &= 0.5 \frac{2 - (e^{-a} e^{1b} + e^a e^{-1b}) z^{-1}}{1 - e^{1b} z^{-1} - e^{-a} e^{1b} z^{-1} + e^{a1} z^{-2}} \\
 &= 0.5 \frac{2 - (e^{-a} e^{1b} + e^a e^{-1b}) z^{-1}}{1 - e^{1b} z^{-1} + e^{-a1} z^{-2}} \\
 &= \frac{1 - e^{-a1} (e^{1b} + e^{-1b}) z^{-1}}{1 - e^{1b} z^{-1} + e^{-a1} z^{-2}}
 \end{aligned}$$

Q5) Solution

$$H(s) = \frac{b}{(s+a)q_1 s^2}$$

$$= \frac{b}{(s+a)(b)} (s+a)(b) = \frac{P_1}{(s+a)(b)} + \frac{P_2}{(s+a)(b)}$$

$$\Rightarrow P_1 = \frac{-1}{a} \quad \text{and} \quad P_2 = \frac{1}{a}$$

So, $H(s) = \frac{1/a}{(s+a)(b)} + \frac{1/a}{(s+a)(b)}$

$$h_c(t) = \mathcal{L}^{-1} \left[\frac{-1}{a} e^{-(a+1b)t} + \frac{1}{a} e^{-(a-1b)t} \right] u(t)$$

$$h_p[n] = h_c[nT] = \delta \left[\frac{-1}{a} e^{-(a+1b)nT} + \frac{1}{a} e^{-(a-1b)nT} \right] y[n]$$

we know that, $H(z) = \sum_{n=0}^{\infty} h_p[n] z^{-n}$

$$= \sum_{n=0}^{\infty} \left[\delta \left[\frac{-1}{a} e^{-(a+1b)nT} z^{-n} \right] + \delta \left[\frac{1}{a} e^{-(a-1b)nT} z^{-n} \right] \right]$$

$$= \frac{-1}{a} \sum_{n=0}^{\infty} \left[e^{-(a+1b)nT} z^{-n} \right] + \frac{1}{a} \sum_{n=0}^{\infty} \left[e^{-(a-1b)nT} z^{-n} \right]$$

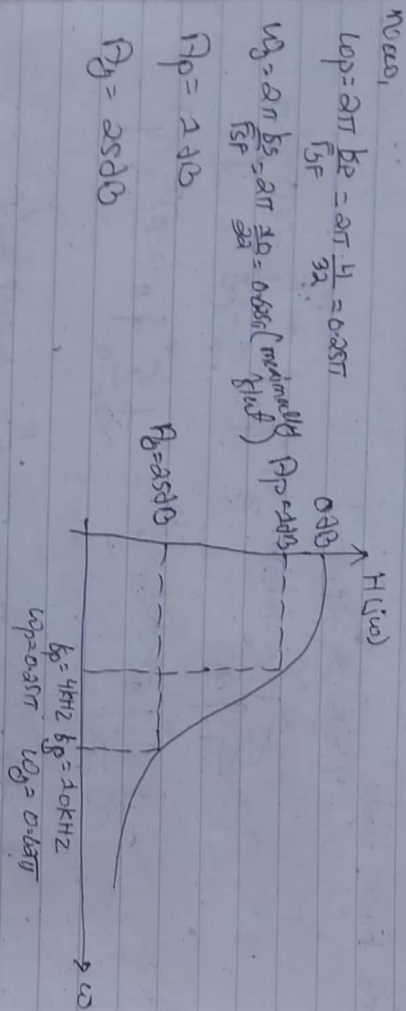
$$= \frac{-1}{a} \left[\frac{1}{1 - e^{-(a+1b)T} z^{-1}} - \frac{1}{1 - e^{(a-1b)T} z^{-1}} \right]$$

$$= \frac{1}{a} \frac{e^{-aT} (e^{1bT} - e^{-1bT}) z^{-1}}{1 - e^{a1} (e^{1bT} z^{-1} + e^{-a1} z^{-2})}$$

$$\therefore H(z) = \frac{e^{-aT} (e^{1bT} z^{-1} + e^{-a1} z^{-2})}{1 - e^{a1} (e^{1bT} z^{-1} + e^{-a1} z^{-2})}$$

(36) $\Rightarrow$ Solution
Here,

Pass band edge frequency (f_p) = 4 kHz
Stop Band attenuation (A_s) = 25 dB
Stop band edge frequency (f_s) = 10 kHz
Sampling frequency (f_s) = 32 kHz



$\therefore$ PBT method is used

i) Calculate Σ_p and Σ_s :

$$\Sigma_p = \frac{2}{\pi} \tan \frac{\omega_p}{2} = \frac{2}{\pi} \tan \frac{0.25\pi}{2} = 0.6503.67$$

$$\Sigma_s = \frac{2}{\pi} \tan \frac{\omega_s}{2} = \frac{2}{\pi} \tan \frac{0.625\pi}{2} = 4.5782.77$$

ii) Order of filter (N):

Since specification are in dB

$$N = \frac{\log \left(\frac{20 \log 10 \frac{A_p}{A_s} - 1}{20 \log 10 \frac{A_p}{A_s} - 1} \right)}{20 \log \left(\frac{f_s}{f_p} \right)}$$

$$\Rightarrow N = 3$$

iii) Cut off frequency (ω_c):
Since ω_c is not given and specification is in dB

$$\omega_c = \frac{\omega_p}{\left(\frac{10^{0.25} - 1}{20 \log 10 \frac{A_p}{A_s} - 1} \right)^{\frac{1}{N}}}$$

$$= \frac{0.25\pi}{\left(\frac{10^{0.25} - 1}{20 \log 10 \frac{A_p}{A_s} - 1} \right)^{\frac{1}{3}}}$$

$$\therefore \omega_c = 33205.38$$

iv) Pole-Pair Position (P_i):

$$P_i = \pm \Sigma_c e^{j(N+1+2i)\frac{\pi}{2N}} \quad ; i = 0, 1, 2$$

For $i = 0$:

$$P_0 = \pm 33205.38 e^{j(3+1+0)\frac{\pi}{6}}$$

$$= \pm 33205.38 e^{j\frac{4\pi}{3}}$$

$$= \pm 33205.38 \left[\cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3} \right]$$

$$= \pm 9 - j16602.7 + j28756.7$$

$$\Rightarrow P_0 = 9 - j16602.7 + j28756.7$$

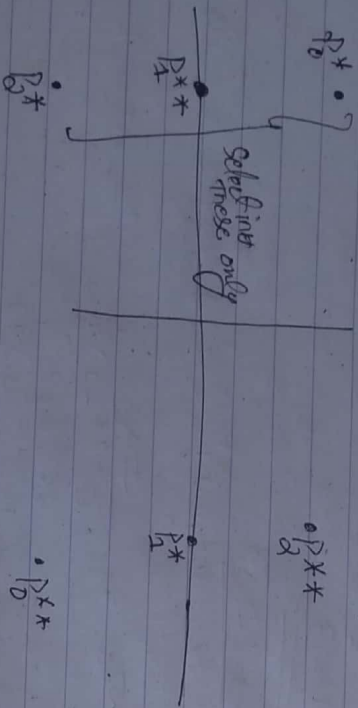
$$+ 16602.7 - j28756.7$$

i=2

$$\Rightarrow P_2 = \left[\begin{matrix} -16602.7 & -28756.7 \\ +16602.7 & +28756.7 \end{matrix} \right] \begin{matrix} P_1^* \\ P_2^* \end{matrix}$$

i=1

$$P_1 = \left[\begin{matrix} 33205.38 & P_1^* \\ -33205.38 & P_2^* \end{matrix} \right]$$



v) Construct function in analog domain: $H(s)$

$$H(s) = \frac{52 \times 10^3}{(s-P_1)(s-P_2)}$$

$$\Rightarrow H(s) = \frac{(33205.38)^2}{(s+33205.38)^2} \left[\frac{s - (-16602.7 - j28756.7)}{s + 91 + j6} \right] \left[\frac{s - (-16602.7 + j28756.7)}{s + 91 - j6} \right]$$

$$\Rightarrow H(s) = \frac{(33205.38)^2}{(s+33205.38)^2} \left[(s+16602.7)^2 + (28756.7)^2 \right]$$

For BJT method,

$$H(z) = H(s) \Big|_{z = \frac{1}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = 64 \times 10^3 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

$$\Rightarrow H(z) = \frac{(33205.38)^2}{\left[64 \times 10^3 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 33205.38 \right]^2} \left[\left(64 \times 10^3 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 16602.7 \right)^2 + (28756.7)^2 \right]$$

$$= \frac{(33205.38)^2}{\dots}$$

Q7) Solution

Pass Band frequency (ω_p) = 0.25π
Stop Band frequency (ω_s) = 0.5π
use here,

$$\Rightarrow \omega_p = 0.25\pi$$

$$\omega_s = 0.5\pi$$

$$\text{Sampling frequency } (f_s) = 1 \text{ kHz}$$



ii) Calculation of ω_p and ω_s is used on

$$\omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right) = 2f_s \tan\left(\frac{\omega_p}{2}\right) = 0.828$$

$$\omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right) = 2f_s \tan\left(\frac{\omega_s}{2}\right) = 2.663$$

iii) Order of Filter:

Since, specification are in dB, we have

$$N = \frac{\log\left(\frac{10^{11.3} - 1}{10^{11.3} - 1}\right)}{2 \log\left(\frac{52}{50}\right)}$$

$$\Rightarrow N = 2.03 \approx 3$$

iii) (at all frequency (ω_c)):

$$\omega_c = \frac{\omega_p}{(10^{11.3} - 1)^{1/N}} = \frac{0.828}{(10^{11.3} - 1)^{1/6}}$$

$$\Rightarrow \omega_c = 2.039$$

iv) Pole-Zero Position (P_i):

$$P_i = \pm \omega_c e^{j(2i-1)\pi/N}, \quad i = 0, 1, 2$$

$$P_0 = \pm 1.039 e^{j(3+1)\pi/6}$$

$$= \pm 1.039 e^{j(2\pi/3)}$$

$$\Rightarrow P_0 = \begin{cases} -0.5195 + j0.895 \\ +0.5195 - j0.895 \end{cases}$$

For $i=2$

$$\Rightarrow P_2 = \begin{cases} -0.5195 - j0.895 \\ +0.5195 + j0.895 \end{cases}$$

For $i=1$

$$P_1 = \begin{cases} -1.039 \\ 1.039 \end{cases}$$

v) Calculating $H_0(s)$:

$$H_0(s) =$$

$$\frac{52^N}{(s + 1.039)^3} \left[(s + 0.5195)^2 + (0.895)^2 \right]$$

Now,

$$H(z) = H_0(s)$$

$$s = 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

$$\Rightarrow H(z) =$$

$$\frac{(1.039)^3}{2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 1.039} \left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 0.895^2 \right]$$

(8) Solution

Here,

$$H(z) = \frac{\omega_c^2}{2 - 0.6z^{-1} + 0.2z^{-2}}$$

Now, for converting to HPF with (LHP) $\omega_c = 0.307\pi$ we use

$$H(z)_{HPF} = H(z)_{LPF}$$

$$z^{-1} = \frac{z^{-1} - a}{1 + az^{-1}}$$

where,

$$a = \frac{\cos(\frac{\omega_p + \omega_s}{2})}{\cos(\frac{\omega_p - \omega_s}{2})}$$

$$= \frac{\cos(\frac{0.3575\pi + 0.3573\pi}{2})}{\cos(\frac{0.3575\pi - 0.3573\pi}{2})}$$

$$\Rightarrow a = 0.576$$

Hence,

$$H(z)_{HPF} = \frac{0.2 + 0.4 \left[\frac{z^{-1} - 0.576}{1 + 0.576z^{-1}} \right]}{1 - 0.6 \left[\frac{z^{-1} - 0.576}{1 + 0.576z^{-1}} \right] + 0.2 \left[\frac{z^{-1} - 0.576}{1 + 0.576z^{-1}} \right]}$$

Q9) Solution

Hence,

Pass-band frequency (ω_p) = $2\pi \times 200 \text{ Hz}$

Stop-band frequency (ω_s) = $2\pi \times 500 \text{ Hz}$

Since, gain = - Attenuation,

$$\text{we get, } A_p = 50 \text{ dB}$$

$$A_s = 120 \text{ dB}$$

$$\text{Sampling frequency (F}_s\text{)} = 5 \text{ kHz}$$

$$\text{Now, } \omega_p = 2\pi \times 200 = 2\pi \times \frac{200}{5000} = 0.08\pi$$

$$\omega_s = 2\pi \times 500 = 2\pi \times \frac{500}{5000} = 0.2\pi$$

ie. IIM method is used.

i) Calculation of ω_s and ω_p :

$$\omega_s = \frac{\omega_s}{T} = 0.2\pi \times 5000 = 1000\pi$$

$$\omega_p = \frac{\omega_p}{T} = 0.08\pi \times 5000 = 400\pi$$

ii) Order (N):

$$N = \frac{\log\left(\frac{10^{0.1A_p} - 1}{10^{0.1A_s} - 1}\right)}{2 \log\left(\frac{\omega_s}{\omega_p}\right)}$$

$$\Rightarrow N = 1.051$$

iii) Cut-off frequency (ω_c):

$$\omega_c = \frac{\omega_p}{(10^{0.1A_p} - 1)^{1/2N}} = \frac{400\pi}{(10^{0.1 \times 50} - 1)^{1/4}}$$

$$\Rightarrow \omega_c = 1036.23$$

ii) Pole-Position (P_i):

$$P_i = 1 \pm \frac{j(N+1+2i)}{2N} \quad ; \quad i = 0, 1$$

For $i=0$:

$$P_0 = \pm 1036.25 e^{j(2+1)\frac{\pi}{4}}$$

$$= \pm 1036.25 e^{j(\frac{3\pi}{4})}$$

$$\Rightarrow P_0 = \int_{732.767}^{-732.767} + \int_{732.767}^{732.767} - j 732.767$$

For $i=1$

$$\Rightarrow P_1 = \int_{732.767}^{-732.767} - j 732.767 + \int_{732.767}^{732.767} + j 732.767$$

v) Calculating $H(s)$:

$$H(s) = \frac{5Z_c}{(s + 732.767)^2} + (732.767)^2$$

$$= \frac{(1036.25)^2}{(s + 732.767)^2} + (732.767)^2$$

$$\frac{b}{(s + 10)^2 + b^2}$$

$$s = -\frac{1}{T}$$

$$New_{01} = \frac{(1036.25)^2}{(732.767)^2} \times \frac{(732.767)^2}{(s + 732.767)^2 + (732.767)^2} \quad \left(\frac{b}{s^2 + 2s + b^2} \right)$$

Using ITM formula,

$$H(z) = \left(\frac{1036.25^2}{732.767^2} \right) \times \frac{e^{-732.767T} \sin\left(\frac{732.767T}{T}\right) z^{-1}}{1 - 2e^{-732.767T} \cos(732.767T) z^{-1} + e^{-2 \times 732.767T} z^{-2}}$$

$$\left(\text{put calculated value in mod } \right) = \frac{1416.853}{1 - 1.727z^{-1} + 1.491z^{-2}}$$

$$\Rightarrow H(z) =$$

$$= \frac{1 - \frac{3.228e^{-188.498T}}{98.457} + \frac{12.496T}{98.457}}{1 - 1.708z^{-1} + 0.742z^{-2}}$$

8.10) Solution Here,

Order (N) = 2

Cut-off frequency (b_c) = 1 kHz $\Rightarrow \omega_c = 2\pi b_c = 0.2\pi$

Sampling frequency (f_{sf}) = 10⁴ Hz

"BPT Method"

$$\text{cut-off frequency (} \omega_c \text{)} = \frac{b_c}{T} \text{ rad/sec}$$

$$= 2f_{sf} \sin\left(\frac{0.2\pi}{2}\right)$$

$$\Rightarrow \omega_c = 6498.39$$

Now, Pole-Pair Position (P_i) = $\pm \omega_c e^{j(N+1+2i)\frac{\pi}{2N}}$; $i=0,1$

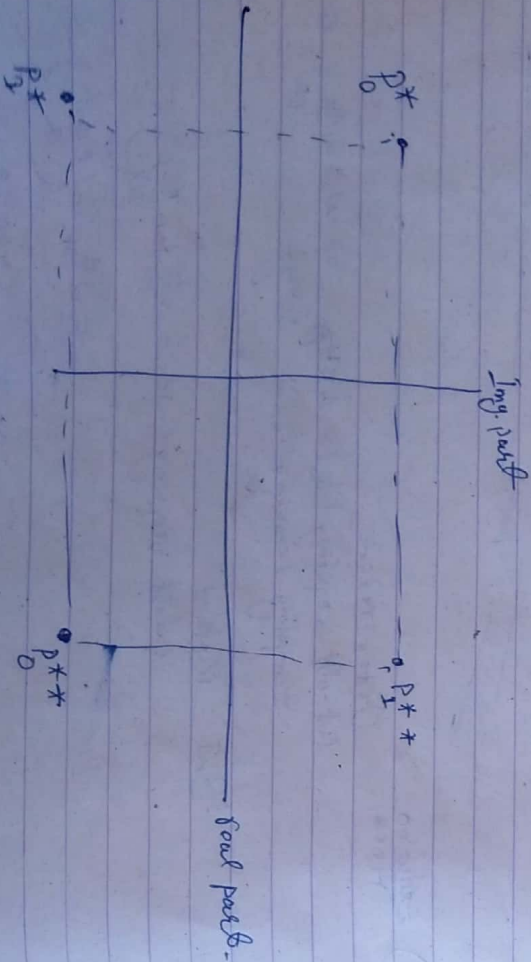
For $i=0$

$$P_0 = \pm \omega_c e^{j(2+2)\frac{\pi}{2}} = \pm \omega_c e^{j(2\pi)} = \pm 6498.39 \left[\cos\left(\frac{3\pi}{4}\right) + j \sin\left(\frac{3\pi}{4}\right) \right]$$

$$\Rightarrow P_0 = \begin{cases} -4595.05 + j40595.05 & P_0^* \\ 4595.05 - j40595.05 & P_0^{**} \end{cases}$$

For $i=1$

$$\Rightarrow P_1 = \begin{cases} -4595.05 - j40595.05 & P_1^* \\ 4595.05 + j40595.05 & P_1^{**} \end{cases}$$



Now, $H_a(s) = \frac{(6498.39)^2}{(s^2 + 4595.05s^2 + (4595.05)^2)}$

$$= \frac{(6498.39)^2}{4595.05} \times \frac{4595.05}{(s^2 + 4595.05s^2 + (4595.05)^2)}$$

Here, $a = 4595.05$; $b = 4595.05$; $T = \frac{1}{2000} \text{ sec}$
So, Using formula, we get,

$$H(z) = \frac{0.29022 \times e^{-4595.05 \times 10^{-4}} \sin(4595.05 \times 10^{-4}) z^{-1}}{1 - 2e^{-4595.05 \times 10^{-4}} \cos(4595.05 \times 10^{-4}) z^{-1} + e^{-2 \times 4595.05 \times 10^{-4}}}$$

$$\Rightarrow H(z) = \frac{0.46714 z^{-1}}{1 - 1.2022 z^{-1} + 0.2285 z^{-2}}$$

Q2a) Solution
Here,

Pass Band frequency (ω_p) = 0.35π

Stop Band frequency (ω_s) = 0.65π

pass band ripple (δ_p) = 0.25

$\Rightarrow \omega_p = -20 \log(1 - \delta_p) = 1.42, 0.13$

Stop band ripple (δ_s) = 0.25

$\Rightarrow \omega_s = -20 \log(\delta_s) = 12.042, 0.13$

Now, GLT method is used on

$\omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right) = \frac{2}{T} \tan\left(\frac{0.35\pi}{2}\right) = \underline{\underline{2.22}}$

$\omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right) = \frac{2}{T} \tan\left(\frac{0.55\pi}{2}\right) = \underline{\underline{2.34}}$

i) Order (N):

$$N = \frac{\log \left(\frac{10^{0.1A_s} - 1}{10^{0.1A_p} - 1} \right)}{2 \log \left(\frac{\omega_c}{\omega_p} \right)}$$

$$\Rightarrow N = 0.81 = 3$$

ii) Cut-off frequency (ω_c):

$$\omega_c = \frac{\omega_p}{(10^{0.1A_p} - 1)^{1/N}}$$

$$\Rightarrow \omega_c = 1.431$$

iii) Pole Position (p_i):

$$p_i = \pm \omega_c e^{j(\omega_c + 2\pi i) \frac{1}{2N}}, \quad i = 0, 1, 2$$

$$\text{For } i=0 \Rightarrow p_0 = \pm 1.431 e^{j(\frac{\pi}{3})}$$

$$= \pm 1.431 \left[\cos \frac{\pi}{3} + j \sin \frac{\pi}{3} \right]$$

$$\Rightarrow p_0 = -0.7155 + j1.2392 \quad p_0^*$$

$$0.7155 - j1.2392 \quad p_0^{**}$$

For $i=1$

$$p_1 = 1.431 \quad p_1^*$$

$$-1.431 \quad p_1^{**}$$

For $i=2$

$$p_2 = -0.7155 - j1.2392 \quad p_2^*$$

$$0.7155 + j1.2392 \quad p_2^{**}$$

iv) $H(s)$:

$$H(s) = \frac{(1.431)^3}{(s + 1.431) [(s + 0.7155)^2 + (1.2392)^2]}$$

v) $H(z)$:

Now,

$$H(z) = H(s) \Big|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} = 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

$$\Rightarrow H(z) = \frac{2.93}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 1.431 \right] \left[\left(2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.7155 \right)^2 + 1.5356 \right]}$$

Q22) Solution

Here,

$$1.94$$

$$-3.05810 \leq |H(e^{j\omega})| \leq 0$$

$$|H(e^{j\omega})| \leq 1 \quad 0 \leq \omega \leq 0.2\pi$$

Here,

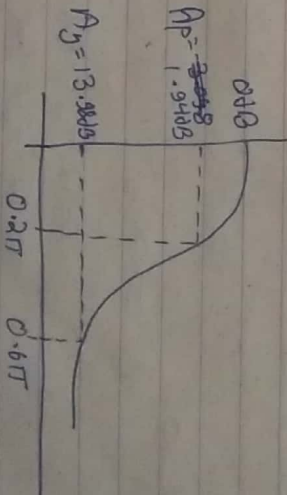
$$\omega_p = 1.9410$$

$$\omega_{sp} = 0.2\pi$$

$$\omega_{\omega} = 13.9810$$

$$T = 2\pi$$

"Using BAT method"



i) Calculating $\angle \Phi \angle \angle S$:

$$\angle Z_P = \frac{2}{\pi} \tan^{-1} \frac{\omega}{2} = 0.649$$

$$\angle S = \frac{2}{\pi} \tan^{-1} \frac{\omega}{2} = 2.752$$

ii) Order (N):

$$N = \frac{10 \left(\frac{20^{0.1A_s} - 1}{20^{0.1A_p} - 1} \right)}{2 \log \left(\frac{\omega_s}{\omega_p} \right)}$$

$$\Rightarrow N = 1.23 = 2$$

iii) Cutoff frequency (ω_c):

$$\omega_c = \frac{\omega_p}{(20^{0.2A_p} - 1)^{1/2N}}$$

$$\Rightarrow \omega_c = 0.749$$

iv) Pole-Pair Position (P_1):

$$P_1 = \pm \omega_c e^{j(N+2+2i)\frac{\pi}{2N}}; i=0,1$$

$$\text{For } i=0 \quad P_0 = \pm 0.749 e^{j(a+1)\frac{\pi}{4}} = \pm 0.749 e^{j(\frac{3\pi}{4})}$$

$$\Rightarrow P_0 = \begin{cases} -0.52 + j0.52 & P_0^* \\ 0.52 - j0.52 & P_0^{**} \end{cases}$$

$$\text{For } i=1 \quad P_2 = \begin{cases} -0.52 - j0.52 & P_2^* \\ 0.52 + j0.52 & P_2^{**} \end{cases}$$

v) $H(s)$:

$$H(s) = \frac{(0.749)^2}{[(s+0.52)^2 + (0.52)^2]}$$

vi) $H(z)$:

Using BLT method,

$$H(z) = H(s) \Big|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$\Rightarrow H(z) = \frac{0.56}{[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.52]^2 + 0.2704}$$

Q14) Solution Here,

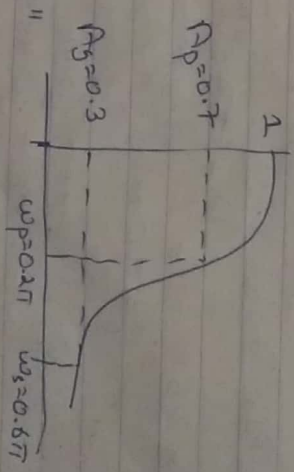
$$0.7 \leq |H(e^{j\omega})| \leq 1 \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.3 \quad 0.6\pi \leq \omega \leq \pi$$

We get,

$$\begin{aligned} \omega_p &= 0.7 \\ \omega_s &= 0.2\pi \\ \omega_g &= 0.3 \\ T &= 2 \text{ sec} \end{aligned}$$

"Imag method is used"



1) Calculating ω_p and ω_s :

$$\omega_p = \frac{\omega_p}{T} = 0.2\pi$$

$$\omega_s = \frac{\omega_s}{T} = 0.6\pi$$

2) Order (N):

$$N = \frac{\log\left(\frac{1/A_p^2 - 1}{1/A_s^2 - 1}\right)}{2 \log\left(\frac{\omega_p}{\omega_s}\right)}$$

$$\Rightarrow N = 1.034 \approx 2$$

3) Anti-alias frequency (ω_c):

$$\omega_c = \frac{\omega_p}{\left(\frac{1}{A_p} - 1\right)^{1/N}}$$

$$\Rightarrow \omega_c = 0.622$$

4) Pole-Zero Position (P_i):

$$P_i = \pm \omega_c e^{j(\frac{3\pi}{4})} \quad i=0,1$$

For $i=0$:

$$P_0 = \pm 0.622 e^{j(\frac{3\pi}{4})}$$

$$\Rightarrow P_0 = \begin{matrix} -0.44 + j0.44 \\ 0.44 - j0.44 \end{matrix} \quad \begin{matrix} P_0^* \\ P_0^* \end{matrix}$$

For $i=1$:

$$P_1 = \begin{matrix} -0.44 - j0.44 \\ 0.44 + j0.44 \end{matrix} \quad \begin{matrix} P_1^* \\ P_1^* \end{matrix}$$

1) $H(s)$:

$$H(s) = \frac{(0.622)^2}{[(s+0.44)^2 + (0.44)^2]}$$

2) $H(z)$:

$$H(s) = \frac{0.386}{0.44} \times \frac{0.44}{[(s+0.44)^2 + (0.44)^2]} \quad \left\{ \frac{b}{(s+a)^2 + b^2} \right\}$$

Using IIR method,

$$H(z) = \frac{0.386 e^{-0.44T}}{1 - 2e^{-0.44T} \cos(0.44\pi T) z^{-1} + e^{-2 \times 0.44T} z^{-2}}$$

$$\Rightarrow H(z) = \frac{4.384 \times 10^{-3}}{1 - 1.16 z^{-1} + 0.414 z^{-2}}$$

Now,

$$H(z)_{LPF} = H(z)_{LPF} \quad \text{with } \omega_p = 0.4\pi$$

$$\omega_p = 0.4\pi \quad \text{Now, } a = \frac{\sin\left(\frac{0.4\pi \times 0.4\pi}{2}\right)}{\sin\left(\frac{0.4\pi + 0.4\pi}{2}\right)}$$

$$\Rightarrow a = -0.44$$

$$\Rightarrow H(z)_{LPF} = \frac{0.24 \times \left(\frac{1 - (-0.44)}{1 + 0.44}\right)}{1 - 1.16 \left(\frac{1 - (-0.44)}{1 + 0.44}\right) z^{-1} + 0.414 \left(\frac{1 - (-0.44)}{1 + 0.44}\right)^2 z^{-2}}$$

Q15) Solution

Here, $H_0(s) = \frac{s+0.1}{(s+0.1)^2+16}$; So $H_0(s)$ has resonant frequency, $\omega_0 = \sqrt{16} = 4$

Resonant frequency (ω_0) = $\frac{\omega}{2}$

We have,

$$\omega_0 = \frac{\omega}{2} \tan\left(\frac{\omega_0}{2}\right)$$

$$\Rightarrow 4 = \frac{\omega}{2} \tan\left(\frac{\omega}{4}\right)$$

$$\Rightarrow 4 = \frac{\omega}{2} \tan\left(\frac{\omega}{4}\right)$$

$$\Rightarrow T = \frac{2}{\omega}$$

"BLT method is used"

$$H(z) = H_0(s) \Big|_{s=\frac{1-z^{-1}}{1+z^{-1}}} = 4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

$$\Rightarrow H(z) = \frac{4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.1}{\left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 16}$$

$$= \frac{4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.1}{\left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 16}$$

$$= \frac{(4.1 - 3.9z^{-1})}{(4.1 - 3.9z^{-1})^2 + 16}$$

$$= \frac{(4.1 - 3.9z^{-1})}{(1+z^{-1})^2 + 16(1-z^{-1})^2}$$

$$= \frac{(4.1 - 3.9z^{-1})}{(1+z^{-1})^2 + 16(1-z^{-1})^2}$$

$$= \frac{(4.1 - 3.9z^{-1})}{(1+z^{-1})^2 + 16(1-z^{-1})^2}$$

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$$= \frac{(4.1 - 3.9z^{-1})}{(1+z^{-1})^2 + 16(1-z^{-1})^2}$$

$$= \frac{0.128 + 0.006z^{-1} - 0.128z^{-2}}{1 + 0.0006z^{-1} + 0.9752z^{-2}}$$

$$= \frac{(2+1)(2-0.95)}{(2-0.987e^{j\pi/4})(2-0.987e^{-j\pi/4})}$$

$\therefore$ Poles at: $P_1, P_2 = 0.987e^{\pm j\pi/4}$

Zeros at: $Z_1, Z_2 = -1, 0.95$

Theory

2) Explain about IIR filter design using IM method?
 ⇒ The steps to design IIR filter using IM method are as follows:

i) Calculate Ω_p and Ω_s using ω_p and ω_s

$$\rightarrow \Omega = \frac{\omega}{T}$$

ii) Order of filter is calculated as:

$$N = \frac{\log\left(\frac{10^{0.1A_s} - 1}{10^{0.1A_p} - 1}\right)}{2 \log\left(\frac{\Omega_s}{\Omega_p}\right)} \quad \left\{ \begin{array}{l} \text{specifications in dB} \end{array} \right.$$

$$N = \frac{\log\left(\frac{1/A_s^2 - 1}{1/A_p^2 - 1}\right)}{2 \log\left(\frac{\Omega_s}{\Omega_p}\right)} \quad \left\{ \begin{array}{l} \text{specifications not in dB} \end{array} \right.$$

iii) Cut-off frequency:

↳ If ω_c is given then

$$\Omega_c = \frac{\omega_c}{T}$$

↳ If ω_c is not given

$$i) \Omega_c = \frac{\Omega_p}{(10^{0.1A_p} - 1)^{1/2N}} \quad \left\{ \begin{array}{l} \text{dB scale} \end{array} \right.$$

$$ii) \Omega_c = \frac{\Omega_p}{(1/A_p^2 - 1)^{1/2N}} \quad \left\{ \begin{array}{l} \text{not in dB scale} \end{array} \right.$$

iv) Pole - Pairs ~~Poles~~ Poles (P₁)

$$P_i = \pm s_2 e^{j(\omega + 2\pi i + \pi)} \frac{\pi}{\omega}, \text{ where } i = 0, 1, 2, 3, \dots, (n-1)$$

v) Zeros of function in s-plane [H₀(s)]

$$H_0(s) = \frac{s^2 e^{\pi}}{(s-P_0)(s-P_1)(s-P_2) \dots}$$

ok pole pairs are complex conjugate then,

$$H_0(s) = \frac{s^2 e^{\pi}}{(s-s_1)(s-s_1^*)(s-s_2)(s-s_2^*)}$$

vi) Finally we determine transfer function in digital domain using IM method.

a) $H_0(s) = \frac{1}{s-P_1} \xrightarrow{\text{Mapping}} H(z) = \frac{1}{1 - e^{P_1 T} z^{-1}}$

b) $H_0(s) = \frac{s \pm b}{(s \pm a)^2} \xrightarrow{\text{Mapping}} H(z) = \frac{1 - e^{-aT} \cos bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}}$

c) $H_0(s) = \frac{1}{s^2 + 1} \xrightarrow{\text{Mapping}} H(z) = \frac{e^{-aT} \sin bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}}$

2) Explain about the FIR filter design using BLT method. The steps to calculate design FIR filter using BLT method are described below:

i) Calculate ω_p and ω_s using ω_p and ω_s

$$\omega_c = \frac{\omega_p + \omega_s}{2}$$

ii) Calculate Order of filter (N).

$$N = \frac{1}{2} \frac{\log \left(\frac{10^{0.1 A_s} - 1}{10^{0.1 A_p} - 1} \right)}{\log \left(\frac{\omega_s}{\omega_p} \right)} \quad \left\{ \begin{array}{l} \text{specifications in dB} \\ \text{specifications not in dB} \end{array} \right.$$

$$N = \frac{1}{2} \frac{\log \left(\frac{10^{0.1 A_s} - 1}{10^{0.1 A_p} - 1} \right)}{\log \left(\frac{\omega_s}{\omega_p} \right)} \quad \left\{ \begin{array}{l} \text{specifications not in dB} \\ \text{specifications in dB} \end{array} \right.$$

iii) Calculate frequency:

$$\omega_c = \frac{2 \tan(\frac{\omega_c}{2})}{T} \quad \text{if } \omega_c \text{ is given}$$

iv) ω_c is not given,

$$\omega_c = \frac{\omega_p}{(10^{0.1 A_p} - 1)^{1/2}} \quad \left\{ \begin{array}{l} \text{specifications in dB} \\ \text{specifications not in dB} \end{array} \right.$$

$$\omega_c = \frac{\omega_p}{(10^{0.1 A_p} - 1)^{1/2}} \quad \left\{ \begin{array}{l} \text{specifications not in dB} \\ \text{specifications in dB} \end{array} \right.$$

iv) Pole-Zero Pairs (P_i)

$$P_i = \pm \sum_{k=1}^N e^{j(N+2i+1)\frac{\pi}{2N}}, \text{ where } i=0,1,2,3 \dots (N-1)$$

v) Transfer function in s-domain [H_q(s)]

$$H_q(s) = \frac{\sum_{k=1}^N s^k}{(s-p_0)(s-p_1)(s-p_2) \dots}$$

or Pole-pairs are complex conjugate then,

$$H_q(s) = \frac{\sum_{k=1}^N s^k}{(s-s_1)(s-s_1^*)(s-s_2)(s-s_2^*)}$$

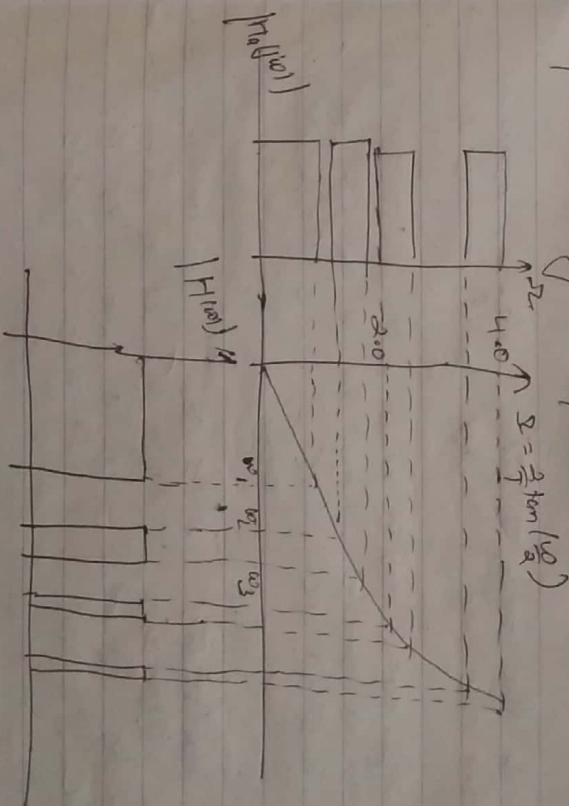
vi) Finally, we determine transfer function in digital domain [H_q(z)] using BHT method.

$$H_q(z) \Big|_{s=\frac{z-1}{z+1}} = H(z)$$

a3) What is frequency warping effect during IIR filter design? The relationship between analog and digital frequencies in bilinear transformation is given by,

$$\omega = \frac{2}{T} \tan\left(\frac{\omega_d T}{2}\right)$$

For smaller values of ω , these are linear relationship between ω and ω_d . But for large values of ω , the relationship is non-linear. This non-linearity or warping effect in frequency axis. This is known as warping effect. This effect compresses the magnitude and phase response at high frequencies.



Q: In bilinear B, warping effect on amplitude response

6) Compare and Contrast:

a) Analog filters and Digital Filters:

Analog Filters	Digital Filters
i) It works on analog signals or actual signal.	i) It operates on the digital samples of the signal.
ii) It is defined by linear differential equations.	ii) It is defined by using linear difference equations.
iii) Elemental components like resistors, inductors, capacitors are used.	iii) adders, subtractors, delays or other logical components are used.
iv) It is stable and causal, this must be a rational function of S , where coefficients are real.	iv) It is stable and causal, this must be a rational function of z , where coefficients are real.
v) For stability and causality, the poles of transfer function should lie on the left side half of S plane.	v) For stability and causality, the poles of transfer function should lie inside the unit circle in z plane.

b) IIR method and FIR method:

IIR method	FIR method
i) Poles are transferred using the expression: $S \rightarrow \frac{1}{1 - e^{Tz}}$	i) Poles are transferred as, $S = \frac{2}{T} \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right)$
ii) Mapping is many to one.	ii) Mapping is one to one.
iii) Aliasing effect is present.	iii) No aliasing effect.
iv) It is not suitable to design all types of filters.	iv) Suitable to design all types of filters.
v) Only poles of the system can be mapped.	v) Poles as well as zeros can be mapped.
vi) No frequency warping effect.	vi) Frequency warping effect is present.

2) Butterworth, Chebyshev and Elliptic filter design method:

Parameters	Butterworth	Chebyshev	Elliptic
Ripples in Pass Band	No	Yes	Yes
Ripples in stop Band	No	No	Yes
Transition B/w pass band and stop band	Widest	Narrower	Narrowest
Order for given filter specification	Highest	Lower	Lowest
Order of filter (in)	$\frac{10 \log(1/\epsilon)}{20 \log(1/\epsilon_p)}$ $\epsilon = \sqrt{10^{0.1 \text{ dB}} - 1}$	$\cosh^{-1}(\frac{1}{\epsilon_p})$ $\cosh^{-1}(\frac{1}{\epsilon_p})$	$\frac{K(\frac{\epsilon_p}{\epsilon_s}) K \sqrt{1 - \frac{\epsilon_p^2}{\epsilon_s^2}}}{K(\frac{\epsilon_p}{\epsilon_s}) K \sqrt{1 - (\frac{\epsilon_p}{\epsilon_s})^2}}$ $g = \sqrt{1 + g^2}$

3) FIR filter and IIR filter

Parameters	FIR filter	IIR filter
Pole-Zero location	All zero system	Pole-zero system or All pole filter
Transfer function	$H(z) = \sum_{k=0}^m b_k z^{-k}$	$H(z) = \frac{\sum_{k=0}^m b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$
Stability	Always stable	Conditionally stable
Efficiency	Low	High
Speed	Slow	Fast
Phase Response	Linear	Non-linear
Unit delay	Doesn't produce	May produce
Impulse and amplitude	More operations	Fewer operations